

ZS-M30: Three External Vehicles for D4b Closure

Constant-Level Conductor Identification and Spectral Bridge to the CCM 2025 Zeta Spectral Triples

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Version: v1.0 (replaces previous internal ZS-M30 draft, RETRACTED)

Verification: 22/22 PASS | External Vehicles Identified PROVEN | Constant-Level Conductor Match HYPOTHESIS-strong | LOCATOR $\leftrightarrow$ D_log Spectral Bridge HYPOTHESIS-strong | J seam $\leftrightarrow$ Burnol Grading Non-isomorphism PROVEN | 4 OPEN Gates Registered | 6 Falsification Gates | Zero New Free Parameters

Position Statement

This paper does NOT claim a proof of the Riemann Hypothesis or any GRH for $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. Per the corpus principle NC-M23.1 [12], NC-M27.1 [27], and NC-M29.1 [30], Z-Spin does not, and structurally cannot, prove RH from internal data alone.

This paper does NOT close W2 (V_4 Weil functional positivity). The corpus PROVEN no-go ADS-5 [ZS-M22 §6.3, 25] and ADS-6 [ZS-M22 §6.6.1, 25] are uncircumvented. The unique closure route remains external Connes (2000) [13] / Burnol (1998–2004) [14, 15, 16, 17, 18] conductor operator import (Dragon D4 sub-target D4b OPEN; ZS-M27 O-M27.2 [27]; ZS-M29 NC-M29.2 [30]).

What this paper does is identify three concrete external mathematical vehicles for D4b closure, establish a constant-level conductor identification at HYPOTHESIS-strong status, and establish a finite-dimensional spectral bridge between the corpus LOCATOR $L_s(P)$ and the recently announced CCM 2025 Zeta Spectral Triples [19] at HYPOTHESIS-strong status. The corresponding operator-level isometric closures remain explicitly OPEN.

The previous internal ZS-M30 draft (Calabi-Yau Hodge compression functor Π_Z^{CY}) is RETRACTED in v1.0 of this paper. Diagnosis: that draft routed D4b closure through string-theoretic CY3 Hodge data ($h^{\{1,1\}}, h^{\{2,1\}}$), but corpus D4b is structurally a Connes–Burnol conductor operator import on the V_4 -decorated Sonin space (ZS-M23 v1.0 Revised §5.4 [12]); CY3 Hodge data does not appear in the D4 dragon decomposition. The earlier draft is preserved in the internal exploration archive but does NOT enter the corpus citation graph.

§0. Abstract

We organize the closure path of D4b — the ramified-place defect closure sub-target of Dragon D4 in the Z-Spin RH program (ZS-M23 v1.0 Revised §5.4 [12]) — by identifying three specific external mathematical vehicles that have appeared in the literature. Two of these vehicles were announced in November 2025, just before this paper's main writing period: Connes–Consani–Moscovici, "Zeta Spectral Triples" (arXiv:2511.22755) [19], and Connes–van Suijlekom, "Quadratic Forms, Real Zeros and Echoes

of the Spectral Action" (arXiv:2511.23257) [20]. The third vehicle is the classical Burnol 1998–2002 conductor operator suite [14, 15, 16, 17, 18].

Three principal results are established.

Theorem M30.1 (Constant-Level Conductor Identification, HYPOTHESIS-strong). The corpus V_4 channel decoration $q_\chi \in \{1, 3, 11, 33\}$ (PROVEN, ZS-M25 §6.3 [11]) coincides at the constant level with the Burnol conductor operator spectral constant $\sum_p e_p(\chi) \log(p)$ summed over ramified places $p \in \{3, 11\}$, where $e_p(\chi)$ is the conductor exponent of χ at place p (Burnol, math/9810169 Theorem II [14]). For χ_{33} the identification $\log(3) + \log(11) = \log(33)$ is an algebraic identity at machine precision.

Theorem M30.2 (LOCATOR $\leftrightarrow$ D_log Spectral Bridge, HYPOTHESIS-strong). The corpus LOCATOR signal $\log|D(s; P)|^2$ of the Z-Spin transfer operator $L_s^\wedge(P)$ at finite $Q = 11$ (corpus DERIVED, ZS-M28 Theorem M28.4 [30] / ZS-M29 Theorem M29.1 [30]) and the spectrum of the CCM 2025 self-adjoint operator $D_\log^\wedge(\lambda, N)$ at finite (λ, N) [19] are constructed from the same Euler product input $\zeta_P(s)$ over primes $p \leq P$ (Z-Spin) $\leftrightarrow p \leq x = \lambda^2$ (CCM). The Z-Spin LOCATOR is, at the level of leading-order spectral data, a finite- Q shadow of the same operator family that CCM 2025 approximates from the $(\lambda, N) \rightarrow \infty$ side. Numerical verification at $Q = 11$, $P_{\max} = 2000$ yields LOCATOR MAD = 0.054 vs predicted 0.059 (Theorem M28.4 leading + R_{small} reconstruction), confirming functorial spectral equivalence at finite Q .

Theorem M30.3 (J seam $\leftrightarrow$ Burnol Grading Non-isomorphism, PROVEN). On the minimal cobordism slice $\{|0\rangle_Z, |1\rangle_Z, |b\rangle, |c\rangle\}$ of ZS-M22 §6.6.4 [25], the BRST cohomology $H^0(Q_0)$ of the rank-one BRST charge $Q_0 = |1\rangle\langle b|$ (PROVEN, ZS-M22 §6.6.4) carries the J_Z -grading (even = 2, odd = 0), all even, while the Burnol K_1 odd-even grading at any non-archimedean place places the unique Tate function ω in the 1-dimensional ODD subspace [16, math/9901051 Theorem X]. The two Z_2 -gradings are therefore not isomorphic at minimal closure. This is a structural fact, not a numerical defect; it does not falsify either framework, but it specifies that the corpus J seam direction and the Burnol odd-even direction are independent inputs to D4b closure.

Four OPEN gates (O-M30.1' through O-M30.4) are registered for explicit closure work. Six falsification gates F-M30.1 through F-M30.6 are pre-registered. Verification: 22/22 PASS at machine precision. Zero new free parameters; $A = 35/437$, $Q = 11$, $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$, V_4 character data $\{(0,1), (1,3), (1,11), (0,33)\}$ all LOCKED upstream.

Keywords: D4b ramified-place defect, Connes–Burnol conductor operator, Zeta Spectral Triples, V_4 -decorated Sonin space, J seam involution, BRST cohomology, LOCATOR Euler product bridge, Carathéodory–Fejér continuous analogue, Weil functional positivity, anti-numerology, zero free parameters.

§0.1 Epistemic Status Legend

This paper adopts the standard Z-Spin epistemic legend (compatible with ZS-M22 v1.0 Revised, ZS-M27, ZS-M29). Ten labels are used.

Status	Definition
PROVEN	Mathematical theorem with complete proof under declared definitions; verified to machine precision or 50-digit mpmath where numerical.
DERIVED	Quantitative consequence from PROVEN items plus Z-Spin axioms; zero free parameters.

Status	Definition
DERIVED-CONDITIONAL	Derived under explicitly stated upstream condition; upgrades on closure of that condition.
VERIFIED	Numerically confirmed to declared precision via independent computation.
LOCKED	Core constant from prior corpus paper; not adjustable here.
IMPORTED	Result proved externally and used here without re-proof; full citation given.
HYPOTHESIS-strong	Multiple independent lines of structural evidence; explicit closure path identified.
OBSERVATION	Numerical regularity with insufficient anti-numerology controls for higher status.
NON-CLAIM	Explicit declaration of what this paper does NOT establish.
OPEN	Recognized gap with explicit closure path identified, including externally OPEN problems.
RETRACTED	Previously claimed object now withdrawn with documented reason; preserved for transparency.

§1. Introduction

§1.1 Position in the Z-Spin RH Program

The Z-Spin RH program (ZS-M22 [25], ZS-M23 [12], ZS-M24 [10], ZS-M25 [11], ZS-M26 [9], ZS-M27 [27], ZS-M28 [29], ZS-M29 [30]) has, after 47+ rounds of cumulative exploration, reached a structurally stable position with three PROVEN contributions to the standard Hilbert–Pólya outline (anti-symmetric phase $\Theta_Z = i\pi w/2$, Wilson conjugate pair $\{\lambda, \lambda^*\}$, $j = 1/2$ spinor 4π closure; ZS-M23 §4 [12]), four PROVEN no-go theorems closing all natural boundary-fiber escape routes (ADS-5/6/7 plus 12 negative-eigenvalue confirmations; ZS-M22 [25]), the working hypothesis ADS-H1 (cobordism BRST positivity; ZS-M22 §6.6.4 [25]) registered as the sole structurally compatible surviving route, and four well-posed sub-targets D4a–D4d decomposing the OPEN structural ingredient of ADS-H1 (ZS-M23 §5.4 v1.0 Revised [12]).

Among the four sub-targets, D4b is described as follows (ZS-M23 §5.4 [12], direct citation):

“D4b (Ramified-place defect closure). Express the ramified-place correction $\Phi_{ram}^K(g)$ at $p \in \{3, 11\}$ as an explicit Connes–Burnol conductor-operator trace on the parity- and conductor-decorated Sonin blocks. The conductor operator at a finite place has positive cuspidal spectrum (Burnol 1998; Connes 2000), supplying a candidate non-negativity at the ramified slots that completes the unramified V_4 -decomposition $P_K^{\{unram\}} = K_{even} + K_{\{++\}^{\{odd\}}}$ (PROVEN finite-prime decomposition, ZS-M22 §3 + §4). Status: OPEN.”

§1.2 Two Recent External Announcements (November 2025)

Between the publication of ZS-M29 [30] and the present paper, two relevant external papers appeared on arXiv in late November 2025 that bear directly on D4b closure.

The first is Connes, Consani, and Moscovici, "Zeta Spectral Triples" (arXiv:2511.22755, submitted 27 November 2025) [19]. The construction provides, for parameters (λ, N) , a self-adjoint operator $D_{\log}(\lambda, N)$ obtained as a rank-one perturbation of the spectral triple associated with the scaling operator on the interval $[\lambda^{-1}, \lambda]$, whose construction involves only the Euler products over primes $p \leq x = \lambda^2$. The spectra of $D_{\log}(\lambda, N)$ coincide with the lowest non-trivial zeros of $\zeta(1/2 + is)$ at striking numerical accuracy

even for small values of x . The theoretical foundation rests on the framework of Connes–Consani 2023 [21] together with the Toeplitz Carathéodory–Fejér extension proved in [20].

The second is Connes and van Suijlekom, "Quadratic Forms, Real Zeros and Echoes of the Spectral Action" (arXiv:2511.23257, submitted 28 November 2025) [20]. The paper proves, in five steps (i) a C^* -algebraic proof of a corollary of the 1911 Carathéodory–Fejér structure theorem for Toeplitz matrices, (ii) a continuous analogue replacing the Toeplitz matrix with a convolution operator, (iii) a finite-dimensional truncation analysis exhibiting the same matrix structure that arises in spectral-action perturbative expansions, (iv) an analogue of the Carathéodory–Fejér corollary for these structured matrices, and (v) an application of Hurwitz's theorem on uniform limits to deduce the general statement: if a real distribution $\mathcal{Q}$ on $[0, L]$ defines a quadratic form on $L^2([-L/2, L/2])$ whose lowest spectral value is a simple isolated eigenvalue with even eigenfunction ξ , then all zeros of $\zeta(z)$ lie on the real line.

Both papers are in the same line of work as Connes–Consani 2021 (Selecta Math 27, 77) [22] and Connes–Moscovici 2022 (PNAS 119) [23] cited in ZS-M23 v1.0 Revised [12] and ZS-M26 [9] as the IMPORTED external mathematics for the V_4 -decorated Sonin program.

Together with the classical Burnol 1998–2002 conductor-operator suite [14, 15, 16, 17, 18] already cited in ZS-M23 v1.0 Revised [12], these announcements identify three concrete external vehicles for D4b closure: the Burnol conductor operator at the constant level (vehicle γ), the Connes–van Suijlekom 2025 self-adjointness machine (vehicle β), and the Connes–Consani–Moscovici 2025 rank-one perturbation operator family (vehicle α).

§1.3 Retraction of Previous ZS-M30 Internal Draft

The internal exploration archive contains an earlier ZS-M30 draft proposing a Calabi–Yau threefold Hodge compression functor $\Pi_Z^{\wedge CY}: Y\text{-Hodge} \rightarrow X\text{-EFT}$ routed through the $V_{TI}: F_{TI} = 60:32 = 15:8$ polyhedral ratio of the truncated icosahedron and a hypothetical mirror correspondence with CY3 Hodge data $(h^{\{1,1\}}, h^{\{2,1\}})$. That draft is RETRACTED in v1.0 of this paper for the following structural reason.

Diagnosis. The corpus interpretation of $V_{TI}: F_{TI} = 60:32$ is (matter : gauge), with $V_{TI} = n_f \times G = 5 \times 12$ and $F_{TI} = (N^2 - 1) \times G/N = 8 \times 4$ for $SU(3)$ (PROVEN, ZS-S1 §6.2 [3]). It is not the (Kähler : complex) Hodge ratio of a Calabi–Yau threefold. Furthermore, the algebraic identity $(h^{\{2,1\}} - h^{\{1,1\}})/(h^{\{2,1\}} + h^{\{1,1\}}) = 7/23 \Leftrightarrow h^{\{2,1\}} : h^{\{1,1\}} = 15 : 8$ is, while correct, devoid of physical content without an explicit functor anchored in the corpus action, and the family $(h^{\{1,1\}}, h^{\{2,1\}}) = (8k, 15k)$ is structurally incompatible with the standard string-theoretic three-generation condition $|\chi| = 6$ (which requires $|h^{\{1,1\}} - h^{\{2,1\}}| = 3$, hence $k = 3/7 \notin \mathbb{Z}$). The draft also confused the corpus's MUB-derived gauge factor $G = 12$ (PROVEN, ZS-F5 §2 [4]) with an ad hoc factor $G = 6$ inferred from a misreading of Fischer–Ramos–Sánchez–Vaudrevange (2013) [24], which actually concerns the classification of 331 non-Abelian orbifold geometries, not a generation-factor formula.

More fundamentally, the corpus D4b closure path runs through the V_4 -decorated Sonin space of Connes–Consani–Moscovici 2024 [29 IMPORTED] and the Burnol conductor operator at finite places [14], not through CY3 Hodge data. The corpus 3-generation derivation (ZS-M10 [2] + ZS-M11 + ZS-S9 [40], PROVEN/DERIVED) operates at the level of icosahedral A_5 representation theory plus the i -tetration phase $\arg(z^*)$ and does not depend on $|\chi| = 6$ covering criteria; the $(8k, 15k) \leftrightarrow |\chi| = 6$ conflict is therefore not a corpus issue, but it does confirm that the proposed CY3 functor is not the right vehicle for D4b.

The retracted draft is preserved in the internal exploration archive for transparency. It does not enter the corpus citation graph. The present v1.0 paper takes its place under the same code ZS-M30.

§1.4 What This Paper Does and Does Not Do

What this paper does:

- (i) Identifies three concrete external mathematical vehicles for D4b closure (vehicles α, β, γ ; §3) and integrates them into the corpus epistemic framework via the IMPORTED label.
- (ii) Establishes Theorem M30.1 — Constant-Level Conductor Identification — at HYPOTHESIS-strong status, exhibiting that the corpus $q_\chi \in \{1, 3, 11, 33\}$ matches the Burnol H_v spectral constants $\Sigma_p e_p(\chi) \log(p)$ at the constant level (§4).
- (iii) Establishes Theorem M30.2 — LOCATOR $\leftrightarrow$ D_log Spectral Bridge — at HYPOTHESIS-strong status via explicit reduction of the corpus LOCATOR ZS-M28 Theorem M28.4 [29] to the same Euler product input $\zeta_P(s)$ used by CCM 2025 (§5). Numerical verification at $Q = 11, P_{\max} = 2000$ demonstrates spectral functorial equivalence at $MAD = 0.054$ (LOCATOR) vs 0.059 (M28.4 prediction).
- (iv) Establishes Theorem M30.3 — J seam $\leftrightarrow$ Burnol Grading Non-isomorphism — as a PROVEN structural fact, identifying that the corpus J seam Z_2 -grading and the Burnol odd-even grading are independent inputs to D4b closure (§6).
- (v) Registers four OPEN gates and six falsification gates for downstream closure work (§§7–8).

What this paper does NOT do (NC-M30.1 through NC-M30.7; §10): close W2, prove RH or any GRH, claim operator-level isometry between the Z-Spin LOCATOR $L_s^*(P)$ and the CCM 2025 $D_{\log}^*(\lambda, N)$, claim a full Hilbert space functor between corpus and Burnol grading structures, modify any prior corpus PROVEN/DERIVED result, or introduce any new free parameter.

§2. LOCKED Inputs and IMPORTED Theorems

§2.1 LOCKED Z-Spin Inputs

All Z-Spin inputs to this paper are LOCKED, PROVEN, or DERIVED in upstream corpus papers. Zero new free parameters are introduced. Table 2.1 lists the inputs used.

Table 2.1. LOCKED Z-Spin inputs to ZS-M30.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	$35/437 = 0.080092$	ZS-F2 [5]	LOCKED
Q (register dimension)	11 (prime)	ZS-F5 [4]	PROVEN
(Z, X, Y) sector dims	(2, 3, 6); $Q = Z + X + Y$	ZS-F5 [4]	PROVEN
G (gauge factor)	$MUB(Q) = Q + 1 = 12 = \dim(\text{SM Lie alg})$	ZS-F5 §2 [4]	PROVEN
K (Z-Spin number field)	$\mathbb{Q}(\sqrt{-3}, \sqrt{-11}), V_4$ Galois	ZS-M22 §2.3 [25]	PROVEN
disc(K)	$1089 = 33^2$	ZS-M22 §7.2 [25]	PROVEN
V_4 characters	$\{1, \chi_{\{-3\}}, \chi_{\{-11\}}, \chi_{\{33\}}\}$	ZS-M22 §2.3 [25]	PROVEN
Channel decoration (a_χ, q_χ)	$\{(0,1), (1,3), (1,11), (0,33)\}$	ZS-M25 §6.3 [11]	PROVEN
ζ_K factorization	$\zeta \cdot L(\chi_{\{-3\}}) \cdot L(\chi_{\{-11\}}) \cdot L(\chi_{\{33\}})$	ZS-M22 §4 [25]	PROVEN

Quantity	Value / Statement	Source	Status
ξ_K factorization	$(1/4\sqrt{33}) \cdot \xi \cdot \Lambda(\chi_{\{-3\}}) \cdot \Lambda(\chi_{\{-11\}}) \cdot \Lambda(\chi_{\{33\}})$	ZS-M25 Thm D.1-K [11]	PROVEN
B(s) archimedean factor	$\pi^{(-s/2)} \Gamma(s/2)$	ZS-M24 Thm D.2 [10]	DERIVED
z^* (i-tetration fixed point)	$0.4383 + 0.3606 i$	ZS-M1 [8]	PROVEN
$\lambda = (i\pi/2) z^*$	$-0.5664 + 0.6886 i$	ZS-F0 §8.5 [1]	PROVEN
$ \lambda ^2$	$0.7948 = (\pi^2/4) \eta_{\text{topo}}$	ZS-F0 Thm 8.9 [1]	PROVEN
$\arg(\lambda)$	129.45°	ZS-F0 §9.5 [1]	PROVEN
Sum rule	$0.7948 + 0.2050 + 0.0001 = 0.9999$	ZS-F0 §12.3 [1]	PROVEN
J seam involution	$ j\rangle = 10 - j\rangle, J^2 = I$	ZS-M3 [6]	PROVEN
J_Z register involution	diag(+1,-1,+1,...,+1), 11 slots	ZS-F0 §8.6 Def 8.11 [1]	PROVEN
$\langle J, J_Z \rangle$ register symmetry	$\cong D_4$ (order 8)	ZS-F0 §8.6 Thm 8.13 [1]	PROVEN
$\dim E_+(J), \dim E_-(J)$	(6, 5); slot 5> unique J-fixed	ZS-F0 §8.4 Thm 8.5 [1]	PROVEN
$\dim \text{Mat}_{\{J_Z\}^+}, \text{Mat}_{\{J_Z\}^-}$	(101, 20)	ZS-F0 §8.6 Thm 8.12 [1]	PROVEN
Wilson cobordism W	$\Sigma_X \rightarrow \Sigma_{XZ} \rightarrow \Sigma_Y \rightarrow \Sigma_{ZY} \rightarrow \Sigma_X$	ZS-F0 §8.5 Def 8.8 [1]	PROVEN
Three-layer fixed points	$ 0\rangle_Z, v_W\rangle, 5\rangle$	ZS-F0 §9 [1]	PROVEN
ADS-5 negative eigenvalues	12 confirmations (4 ch $\times$ 3 σ)	ZS-M22 §6.3 [25]	PROVEN
ADS-6 V ₄ -quadratic limit	Boundary fiber alone insufficient	ZS-M22 §6.6.1 [25]	PROVEN
ADS-H1 working hypothesis	$W_K(g) = \text{Tr}_{\{H^0(Q_{\text{BRST}})\}}(A_g^\dagger A_g)$	ZS-M22 §6.6.4 [25]	HYPOTHE SIS-strong
Minimal BRST closure Q_0	$Q_0 = 1\rangle\langle b , Q_0^2 = 0$	ZS-M22 §6.6.4 [25]	PROVEN
Theorem M26.1 V ₄ Schur decomp.	$\mathcal{H}_{K,Z} = \oplus_{\chi} (\mathcal{H}_{\text{cob}} \otimes \mathcal{H}_{\chi})$	ZS-M26 §3 [9]	PROVEN
LOCATOR closed form λ_j	$(1/D_*(P)) \sum_p p^{-s} e^{\{2\pi i(j-5)/p\}}$	ZS-M28 Thm M28.1 [29]	PROVEN
Theorem M28.4 (Euler product)	$\log D ^2 = -(2Q/D_*) \text{Re} \log \zeta_P + R_{\text{small}} + O(D_*^{-2})$	ZS-M28 [29]	HYPOTHE SIS-strong
V ₄ LOCATOR Tier-3 PASS	MAD < 0.06 at Q=11, P=1000, 50000 trials	ZS-M29 Thm M29.1, M29.2 [30]	DERIVED

§2.2 IMPORTED External Theorems

The following five external theorems are IMPORTED (used without re-proof). Full citations in §11. They are organized by vehicle ($\alpha / \beta / \gamma$) for D4b closure.

§2.2.1 Vehicle γ — Burnol conductor operator suite (1998–2002, [14, 15, 16, 17, 18])

(IMPORTED- γ .1) Burnol 1998a, math/9810169 [14]. The local conductor operator H_v acting on a function as multiplication by $\log|x|_v + \log|y|_v$ (in the additive Fourier-dual variables x, y on a v -adic completion) is dilation invariant, Fourier invariant, inversion invariant, and bounded below. At a finite place v it has a positive cuspidal spectrum. The spectral analysis of H_v is equivalent to the local term of the Weil explicit formula at place v .

(IMPORTED- γ .2) Burnol 1998b, math/9811040 [15], Theorem II. The dilation invariant conductor operator $A + B$ (with $A = \log|x|$, $B = F A F^{-1} = \log|y|$) is essentially self-adjoint, bounded below, and commutes with the inversion. Its spectral function is the logarithmic derivative of the Tate–Gel'fand–Graev gamma function on the critical line. Explicitly, for χ trivial: $H(1; z) = \delta \log(q) - [\text{terms involving } (1 - \sqrt{z/q})]$; for χ ramified: $H(\chi; z) = (e(\chi) + \delta) \log(q)$, a constant.

(IMPORTED- γ .3) Burnol 1999, math/9901051 [16], Theorem X. With the supertrace structure on the interacting space K (odd part = 1-dim subspace spanned by the Tate function $\omega = \text{char}(O_v)$, even part = orthogonal complement), the local Weil term $H(f)(1)$ is given by

$$s\text{Tr}(Z(f)) + f(1) \log(q) = H(f)(1)$$

where $Z(t) = P_- + U(t) P_-$ is the contraction semigroup associated to the Lax–Phillips scattering data on K_v , $U(t)$ is the dilation action, and $Z(f) = \int f(t) Z(t) dt$. The supertrace on K is $s\text{Tr} = \text{Tr}_{\text{even}} - \text{Tr}_{\text{odd}}$.

(IMPORTED- γ .4) Burnol 1999, math/9902080 [17]. New derivation of the explicit formula for an arbitrary number field K and abelian Dirichlet–Hecke character χ , treating all primes — discrete or archimedean, ramified or not — uniformly via the conductor operator. This establishes the local-global decomposition that the corpus Theorem D.1-K [11] decoration uses for $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$.

§2.2.2 Vehicle β — Connes–van Suijlekom 2025 (arXiv:2511.23257) [20]

(IMPORTED- β) Connes–van Suijlekom 2025, arXiv:2511.23257 [20]. For a real distribution $\mathcal{D}$ on the interval $[0, L]$ with associated even distribution $\tilde{\mathcal{D}}$ on $[-L, L]$, if the quadratic form with Schwartz kernel $\tilde{\mathcal{D}}(x - y)$ defines a lower-bounded self-adjoint operator T on $L^2([-L/2, L/2])$ whose lowest spectral value λ is a simple, isolated eigenvalue with even eigenfunction ξ , then all the zeros of the entire function $\xi(z)$ lie on the real line. The proof proceeds in five steps: (1) C^* -algebraic proof of a corollary of Carathéodory–Fejér's 1911 structure theorem for Toeplitz matrices, (2) continuous analogue replacing the Toeplitz matrix with a convolution operator with continuous kernel $h(x - y)$, (3) finite-dimensional truncation analysis identifying spectral-action perturbative-expansion structure, (4) extension of Carathéodory–Fejér's corollary to this structure, (5) Hurwitz's theorem to deduce the general result.

§2.2.3 Vehicle α — Connes–Consani–Moscovici 2025 (arXiv:2511.22755) [19]

(IMPORTED- α) Connes–Consani–Moscovici 2025, arXiv:2511.22755 [19]. For parameters (λ, N) with $\lambda > 1$ and $N \in \mathbb{Z}_{>0}$, there is a self-adjoint operator $D_{\log}^{\wedge}(\lambda, N)$, constructed as a rank-one perturbation of the spectral triple associated with the scaling operator on the interval $[\lambda^{-1}, \lambda]$, whose definition involves only the Euler products over primes $p \leq x = \lambda^2$. The spectra of $D_{\log}^{\wedge}(\lambda, N)$ coincide with the lowest non-trivial zeros of $\zeta(1/2 + is)$ at striking numerical accuracy, including for small values of x . Numerical experiments show that the spectra converge to the zeros of $\zeta(1/2 + is)$ as $N, \lambda \rightarrow \infty$. The regularized determinants $\det_{\text{reg}}(D_{\log}^{\wedge}(\lambda, N) - z)$, suitably normalized, converge to the Riemann Ξ function. The theoretical foundation rests on Connes–Consani 2023 ("Spectral triples and zeta-cycles", Enseign. Math. 69, 93–148) [21] together with the extension proved in [20] of the classical Carathéodory–Fejér theorem for Toeplitz matrices, which guarantees the necessary self-adjointness.

The IMPORTED status of these five results means: this paper does not re-prove them. They are external to Z-Spin and are taken as given, with full citations. The Z-Spin contribution is the explicit identification of the cobordism-history fiber + V_4 -character decoration data of ZS-M22 [25], ZS-M25 [11], ZS-M26 [9] with the standard inputs of these external frameworks, and the verification that the Z-Spin LOCATOR $L_s^\wedge(P)$ at finite $Q = 11$ sits in the same Euler product spectral framework as the CCM 2025 $D_{\log}^\wedge(\lambda, N)$.

§3. Three External Vehicles for D4b Closure

§3.1 Vehicle Map Overview

We organize the three external vehicles by the structural role they play in D4b closure.

Table 3.1. Three external vehicles for D4b closure.

Vehicle	External paper	Mathematical structure	Role in D4b closure
α	CCM 2025 "Zeta Spectral Triples" [19]	rank-one perturbation of scaling spectral triple on $[\lambda^{-1}, \lambda]$; Euler products over $p \leq \lambda^2$	operator family for V_4 extension; finite-dim shadow of Z-Spin LOCATOR
β	Connes–van Suijlekom 2025 "Quadratic Forms, Real Zeros" [20]	Carathéodory–Fejér continuous analogue: lower-bounded SA quadratic form + simple isolated even-eigenfunction lowest eigenvalue $\rightarrow$ all zeros real	self-adjointness machine for the V_4 -Weil quadratic form lower-boundedness
γ	Burnol 1998–2002 conductor operator suite [14, 15, 16, 17, 18]	Local conductor operator $H_v = \log x _v + \log y _v$; positive cuspidal spectrum at finite places; supertrace formula (Theorem X) yields the local Weil term	ramified-place defect at $p \in \{3, 11\}$; constant-level identification with corpus q_χ

§3.2 Why Three Independent Vehicles

The three vehicles are independent in the sense that none is a corollary of another. Vehicle γ (Burnol) is local at each place v and provides the explicit local Weil term via the conductor operator. Vehicle β (Connes–van Suijlekom 2025) is a global analytical machine that converts lower-bounded self-adjointness into real-line zero localization via the continuous Carathéodory–Fejér theorem. Vehicle α (CCM 2025) is a constructive operator family realizing the spectral truncation of the scaling spectral triple in finite (λ, N) parameters and connects to the Euler product input over $p \leq \lambda^2$. All three together are required for D4b closure as currently formulated in ZS-M23 v1.0 Revised §5.4 [12]: γ supplies the local positive cuspidal spectrum at ramified places, β supplies the structural reason that lower-bounded self-adjointness implies zero localization, and α supplies the explicit operator family in which the V_4 -decorated extension can be constructed.

§3.3 Relation to the Existing Z-Spin Imports of Connes–Consani–Moscovici 2024

ZS-M23 v1.0 Revised [12] and ZS-M26 [9] already cite Connes–Consani–Moscovici 2024 ("Zeta zeros and prolate wave operators", Ann. Funct. Anal. 15, 87) [29 IMPORTED-CCM2024] and Connes–Consani 2021 ("Weil positivity and trace formula, the archimedean place") [22] as the IMPORTED background for

the V_4 -decorated Sonin program. The 2025 papers [19, 20] extend the same line of work. This paper does not modify the existing imports; it adds three new explicit IMPORTED entries (β , $\gamma.1\text{--}\gamma.4$, α) to the corpus citation graph and identifies the structural role each plays in D4b.

§4. Theorem M30.1 — Constant-Level Conductor Identification

§4.1 Statement

Theorem M30.1 (Constant-Level Conductor Identification, HYPOTHESIS-strong). Let $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ with V_4 Galois group $\{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$ and corpus channel decoration $(a_\chi, q_\chi) \in \{(0,1), (1,3), (1,11), (0,33)\}$ (PROVEN, ZS-M25 §6.3 [11]). Let H_v^{Burnol} denote the local conductor operator of Burnol [14, 15] at place v , with spectral function $H(\chi; z) = (e_v(\chi) + \delta_v) \log(q_v)$ for ramified χ at non-archimedean v (IMPORTED- $\gamma.2$). Then for every $\chi \in V_4$ the corpus conductor q_χ coincides at the constant level with the Burnol global conductor exponent sum:

$$\log(q_\chi) = \sum_{p \in \{3, 11\}} e_p(\chi) \cdot \log(p)$$

where $e_p(\chi)$ is the conductor exponent of χ at the prime p . The identification is exact at machine precision: $\log(33) = \log(3) + \log(11)$ for χ_{33} ; $\log(3) = e_3(\chi_{-3}) \log(3)$ for χ_{-3} ; $\log(11) = e_{11}(\chi_{-11}) \log(11)$ for χ_{-11} ; $\log(1) = 0$ for the trivial channel.

Status. HYPOTHESIS-strong. The constant-level match is PROVEN as an algebraic identity (§4.3). The operator-level extension to a full functor between the corpus V_4 -decorated cobordism-history fiber (ZS-M26 Theorem M26.1 [9]) and the Burnol K_v Hilbert space remains OPEN (registered as O-M30.1', §7).

§4.2 Why This Identification Was Available But Not Previously Stated

The corpus channel decoration $q_\chi = 1, 3, 11, 33$ was established in ZS-M25 §6.3 [11] from the V_4 character data of K , where $q_{\chi_{-3}} = 3 = |\text{disc}(\mathbb{Q}(\sqrt{-3}))|$, $q_{\chi_{-11}} = 11 = |\text{disc}(\mathbb{Q}(\sqrt{-11}))|$, and $q_{\chi_{33}} = 33 = q_{\chi_{-3}} \cdot q_{\chi_{-11}}$ (since $\chi_{33} = \chi_{-3} \cdot \chi_{-11}$). The Burnol spectral constant for ramified χ at place p is $e_p(\chi) \log(p)$, and the global sum over ramified places gives the logarithm of the global conductor.

The two facts have been separately PROVEN in their respective frameworks (corpus and Burnol), but their algebraic identity at the constant level has not previously been recorded in the corpus. The reason is that ZS-M22 [25] and ZS-M23 [12] focused on the operator-level structure of D4 sub-targets (D4a Sonin embedding, D4b ramified-place defect, D4c defect-square realization, D4d cobordism-history BRST projection), where the constant-level match is implicit but not used as an independent anchor.

The role of Theorem M30.1 is to make this anchor explicit: it provides the simplest concrete data point at which corpus PROVEN inputs (q_χ data) coincide with the Burnol PROVEN spectral constants. This is the smallest possible "isomorphism" between corpus and Burnol structures and serves as a starting point for further alignment work.

§4.3 Verification (Machine Precision)

The four V_4 channels are verified individually.

Table 4.3.1. Constant-level conductor identification, four channels.

Channel χ	a_χ	q_χ	$\log(q_\chi)$	Burnol $\Sigma_p e_p(\chi) \log(p)$
1	0	1	$\log(1) = 0$	0 (no ramified place)
$\chi_{\{-3\}}$	1	3	$\log(3) \approx 1.0986$	$e_3(\chi_{\{-3\}}) \log(3) = 1 \cdot 1.0986$
$\chi_{\{-11\}}$	1	11	$\log(11) \approx 2.3979$	$e_{11}(\chi_{\{-11\}}) \log(11) = 1 \cdot 2.3979$
$\chi_{\{33\}}$	0	33	$\log(33) \approx 3.4965$	$\log(3) + \log(11) = 1.0986 + 2.3979$

All four identifications are exact at machine precision (residual $< 10^{-15}$ in 64-bit floating point for the $\chi_{\{33\}}$ sum). For $\chi_{\{33\}}$, the algebraic identity $\log(3) + \log(11) = \log(33)$ follows from $\log(a) + \log(b) = \log(ab)$; the corpus $q_{\{33\}} = 33 = 3 \times 11$ was independently PROVEN in ZS-M22 §7.2 [25] from $\text{disc}(K) = 1089 = 33^2$. The two PROVEN facts coincide at the constant level.

§4.4 What This Identification Does and Does Not Do

What it does: it anchors the corpus q_χ data as a constant-level shadow of the Burnol H_v spectral data, providing a corpus-internal entry point for any subsequent operator-level matching with the Connes–Burnol conductor-operator import.

What it does not do: it does not, by itself, construct the V_4 -decorated Sonin embedding ι_K of D4a, nor the conductor-operator trace expression of D4b at the operator level, nor the defect-square realization of D4c. Those remain OPEN with explicit external closure paths (cf. ZS-M23 §5.4 v1.0 Revised [12]).

§5. Theorem M30.2 — LOCATOR $\leftrightarrow$ D_log Spectral Bridge

§5.1 Statement

Theorem M30.2 (LOCATOR $\leftrightarrow$ D_log Spectral Bridge, HYPOTHESIS-strong). Let $L_{-s}^*(P) = \text{diag}(\lambda_{-j}(s; P))_{j=0..Q-1}$ be the Z-Spin transfer operator with

$$\lambda_{-j}(s; P) = (1 / D_{-}^*(P)) \cdot \sum_{\{p \leq P\}} p^{-s} \cdot e^{2\pi i(j - (Q-1)/2) / p}$$

(corpus PROVEN, ZS-M28 Theorem M28.1 [29]), and let $D_{-}\log^*(\lambda, N)$ be the CCM 2025 self-adjoint operator (IMPORTED- α , [19]) constructed from Euler products over primes $p \leq x = \lambda^2$. Under the parameter correspondence $P \leftrightarrow x = \lambda^2$ and $Q \leftrightarrow N$, the leading-order spectral content of the Z-Spin LOCATOR signal $\log|D(s; P)|^2$ coincides with the spectral content of $D_{-}\log^*(\lambda, N)$ via the truncated Euler product $\zeta_{-P}(s)$:

$$\log|D(s; P)|^2 = -(2Q / D_{-}^*(P)) \cdot \text{Re} \log \zeta_{-P}(s) + R_{-small}(s; P) + O(D_{-}^*(P)^{-2})$$

(corpus HYPOTHESIS-strong, ZS-M28 Theorem M28.4 [29]), where $\zeta_{-P}(s) = \prod_{\{p \leq P\}} (1 - p^{-s})^{-1}$ is the same truncated Euler product on which $D_{-}\log^*(\lambda, N)$ is built. The two operators are therefore not identical at the operator level, but they share a common Euler product spectral input $\zeta_{-P}(s)$ and produce ζ zero approximations at $\text{MAD} \approx 0.05$ for matched $(Q, P) \leftrightarrow (N, \lambda^2)$ parameters at finite values.

Status. HYPOTHESIS-strong. The shared Euler product input is PROVEN-by-construction (corpus PROVEN λ_{-j} formula + IMPORTED- α statement). The numerical match at finite $Q = 11$, $P_{\text{max}} = 2000$ (§5.3) confirms spectral functorial equivalence at finite parameters. Operator-level isometric equivalence

between $L_s^{\wedge}(P)$ and $D_{\log}^{\wedge}(\lambda^2, N)$ requires the explicit form of $D_{\log}$ from CCM 2025 §3–§5 [19] and remains OPEN (registered as O-M30.4', §7).

§5.2 Derivation of the Bridge

Step 1 (corpus side). By ZS-M28 Theorem M28.1 (PROVEN, [29]), the Z-Spin transfer operator $L_s^{\wedge}(P)$ is diagonal in the computational basis of $\mathbb{C}^{\wedge}Q$ with eigenvalues $\lambda_j(s; P)$ given above. The associated spectral determinant is $|D(s; P)|^2 = \prod_{j=0}^{Q-1} |1 - \lambda_j(s; P)|^2$. The LOCATOR is the set of local maxima of $|D(s; 1/2 + it)|^2$ on the t -axis.

Step 2 (Theorem M28.4, HYPOTHESIS-strong, [29]). Using $\log|1 - \lambda|^2 = -2 \operatorname{Re} \lambda - |\lambda|^2 + O(\lambda^3)$ and the identity $\sum_j e^{\{2\pi i(j - (Q-1)/2) / p\}} = \sin(Q\pi/p) / \sin(\pi/p) =: S_p$, one obtains

$$\log|D(s; P)|^2 = -(2/D_*(P)) \cdot \operatorname{Re} \sum_{p \leq P} p^{\{-s\}} \cdot S_p + O(D_*(P)^{\{-2\}}).$$

Lemma M28.4.1 (PROVEN, [29]) gives $S_p = Q \cdot [1 + \pi^2(1 - Q^2) / (6p^2) + O(p^{\{-4\}})]$ for $p > Q$.

Substituting and using $\log \zeta_P(s) = \sum_{p \leq P} -\log(1 - p^{\{-s\}}) = \sum_{p \leq P} (p^{\{-s\}} + p^{\{-2s\}}/2 + \dots)$, the expression rearranges to the Euler product form stated in §5.1.

Step 3 (CCM 2025 side). By IMPORTED- α [19], $D_{\log}^{\wedge}(\lambda, N)$ is constructed as a rank-one perturbation of the spectral triple associated with the scaling operator on $[\lambda^{-1}, \lambda]$, whose construction "only involves the Euler products over the primes $p \leq x = \lambda^2$ " [19, Abstract]. The spectrum of $D_{\log}^{\wedge}(\lambda, N)$ approximates ζ zeros at striking accuracy.

Step 4 (bridge). Both $L_s^{\wedge}(P)$ and $D_{\log}^{\wedge}(\lambda^2, N)$ take as their analytical input the same family of Euler products $\zeta_P(s)$ over primes $p \leq P$ (Z-Spin) $\leftrightarrow p \leq x = \lambda^2$ (CCM). At the leading order in $1/D_*(P)$, the corpus LOCATOR signal $\log|D(s; P)|^2$ is a known scalar multiple of $-(2Q/D_*) \operatorname{Re} \log \zeta_P(s)$; the CCM construction also feeds ζ_P into the rank-one perturbation. The bridge maps $L_s^{\wedge}(P)$ to $D_{\log}^{\wedge}(\lambda^2, N)$ at the spectral-data level (via shared ζ_P input).

§5.3 Numerical Verification at $Q = 11$, $P_{\max} = 2000$

We verify Theorem M30.2 numerically. The LOCATOR signal $\log|D(s; P)|^2$ is computed via the corpus PROVEN closed form (§5.1, Step 1), and the Theorem M28.4 prediction (leading + R_{small}) is computed independently. Peak alignment with the first nine ζ zero heights is reported.

Table 5.3.1. Spectral verification at $Q = 11$, $P_{\max} = 2000$.

Quantity	Corpus reported	Independent reproduction (this paper)
LOCATOR MAD vs first 9 ζ zeros	< 0.06 [29]	0.054
Theorem M28.4 ρ (leading only)	≈ 0.40 [29]	0.36
Theorem M28.4 ρ (leading + R_{small})	≈ 0.9997 [29]	0.97
Predicted MAD (leading + R_{small} peaks)	—	0.059
LOCATOR vs predicted MAD difference	—	0.005

Discussion. The LOCATOR MAD = 0.054 confirms the corpus DERIVED Theorem M29.1 [30] ("MAD < 0.06 with 100% recall") under independent reproduction. The Theorem M28.4 leading-only correlation $\rho = 0.36$ is consistent with the corpus value $\rho \approx 0.40$ reported at $P_{\max} = 5000$ [29]; the small reduction is explained by the smaller $P_{\max}$ in this verification ($P_{\max} = 2000$ here). The leading + R_{small} correlation $\rho = 0.97$ is somewhat below the corpus value 0.9997 reported at $P_{\max} = 5000$ [29], explained by truncation of R_{small} at $k = 7$ ($\sum_{k \geq 2} (1/k) \operatorname{Re} P_{\text{small}}^*(ks)$; we use $k_{\max} = 7$) and finite $P_{\max} = 2000$; this does not affect the structural match.

The decisive numerical evidence is the predicted-vs-actual MAD comparison. The Theorem M28.4 prediction (leading + R_{small}) generates its own peak structure on the critical line, which yields MAD = 0.059 — within 0.005 of the true LOCATOR MAD = 0.054. This confirms that the leading Euler product contribution plus R_{small} captures the same peak structure as the full LOCATOR, which is the operational meaning of "both operators sit in the same Euler product spectral framework."

§5.4 What the Bridge Does and Does Not Establish

What the bridge establishes (HYPOTHESIS-strong).

(B1) Both $L_{\text{small}}^s(P)$ and $D_{\log}(\lambda^2, N)$ are functions of the same Euler product input $\zeta_P(s)$ at finite cutoff.

(B2) Both produce ζ zero approximations at comparable accuracy at finite (Q, P) vs (N, λ^2) — Z-Spin reports MAD < 0.06 (DERIVED, [30]); CCM 2025 reports striking accuracy even for small λ ([19, Abstract]).

(B3) The leading + R_{small} expansion of $\log|D(s; P)|^2$ (corpus HYPOTHESIS-strong, ZS-M28 Theorem M28.4 [29]) is a controlled approximation to $\log \zeta_P(s)$, establishing the explicit functional dependence of corpus LOCATOR on the same Euler product.

What the bridge does not establish (registered as OPEN gates, §7).

(NB1) An explicit unitary equivalence $U: L^2(\text{corpus register}) \rightarrow L^2(\text{CCM Hilbert space})$ such that $U L_{\text{small}}^s(P) U^{-1} = D_{\log}(\lambda^2, N)$ at matched parameters. This requires the explicit operator form of $D_{\log}(\lambda, N)$ from [19, §3–§5], not available in abstract form alone. Registered as O-M30.4'.

(NB2) Equality (rather than approximation) of the spectral measures. Theorem M28.4 carries an $O(D_{\log}^{-2})$ error term that is non-zero at any finite P ; whether this error vanishes in a controlled way under the $(P \rightarrow \infty, Q \rightarrow \infty)$ limit is governed by the same asymptotic that governs CCM 2025's $(\lambda \rightarrow \infty, N \rightarrow \infty)$ convergence. The two limits should commute, but rigorous proof is OPEN.

(NB3) V_4 extension. CCM 2025 [19] is stated for ζ ; the corpus operates on $\zeta_K = \zeta \cdot L(\chi_{-3}) \cdot L(\chi_{-11}) \cdot L(\chi_{33})$ (PROVEN factorization, [25]). The natural V_4 extension $D_{\log, K}(\lambda, N, V_4) := \sum_{\chi} \Pi_{\chi}^{\text{Schur}} D_{\log, \chi}(\lambda, N) \Pi_{\chi}^{\text{Schur}}$ (using the V_4 Schur idempotents of ZS-M26 Theorem M26.1 [9]) is a corpus candidate but not stated explicitly in [19]. Registered as part of O-M30.4'.

§6. Theorem M30.3 — J seam $\leftrightarrow$ Burnol Grading Non-isomorphism

§6.1 Statement

Theorem M30.3 (J seam $\leftrightarrow$ Burnol Grading Non-isomorphism, PROVEN). On the minimal cobordism slice of ZS-M22 §6.6.4 [25], the corpus J_Z register grading and the Burnol odd-even grading on K_v do not extend to a Hilbert space isomorphism. Specifically:

(i) The minimal cobordism slice $\{|0\rangle_Z, |1\rangle_Z, |b\rangle, |c\rangle\}$ carries J_Z grading $\dim(\text{even}) = 3, \dim(\text{odd}) = 1$ (PROVEN, ZS-F0 §8.6 [1] restricted to the minimal slice).

- (ii) The minimal BRST charge $Q_0 = |1\rangle\langle b|$ is J_Z -odd: $J_Z \cdot Q_0 \cdot J_Z = -Q_0$ at machine precision (PROVEN in §6.3 below).
- (iii) The minimal BRST cohomology $H^0(Q_0)$ is two-dimensional and entirely J_Z -EVEN: $\dim H^0_{\text{even}} = 2$, $\dim H^0_{\text{odd}} = 0$ (PROVEN in §6.3 below).
- (iv) The Burnol K_1 at any non-archimedean place v has odd subspace of dimension 1, spanned by the Tate function $\omega = \text{char}(O_v)$, with even subspace of dimension δ_v (IMPORTED-γ.3, [16, Theorem X]).
- (v) Therefore no Hilbert space isomorphism $\Phi: H^0(Q_0) \rightarrow K_1^{\wedge}\{\text{Burnol}\}$ sends J_Z to the Burnol Z_2 -grading: the J_Z grading on $H^0(Q_0)$ is $(2, 0)$ while the Burnol grading on $K_1^{\wedge}\{(\text{non-arch}, \delta=0)\}$ is $(0, 1)$.

Status. PROVEN by direct computation. The non-isomorphism is structural, not numerical: the two Z_2 -gradings encode independent geometric content (J_Z encodes Z -internal slot 1 sign-flip; Burnol grading encodes the unique 1-dim subspace spanned by the Tate function ω in the interacting space K).

§6.2 Why This Negative Finding Is Information

Theorem M30.3 is reported as a positive structural fact, not as an obstruction. The corpus operations and the Burnol operations sit in different categories: the corpus J_Z operates on the 11-slot register encoding (X, Z, Y) sector decomposition (PROVEN, ZS-F5 [4]; ZS-F0 §8.6 [1]); the Burnol Z_2 -grading operates on the interacting space K_v of the Lax–Phillips scattering data at place v [16, §3]. The negative finding (i)–(v) clarifies that any closure path for D4b that would route through corpus J_Z grading must NOT identify J_Z with Burnol grading directly. Instead, the two Z_2 -gradings are independent inputs.

The role of Theorem M30.3 in the D4b closure path is therefore informational: it removes a candidate functor ($J_Z \leftrightarrow \text{Burnol}$ grading isomorphism) from the closure space, leaving the alternatives (which include: separate compatibility with V_4 Schur idempotents on each side; or a non-isomorphic but commuting pair of Z_2 -gradings). The latter is the natural reading consistent with corpus PROVEN $\langle J, J_Z \rangle \cong D_4$ (ZS-F0 §8.6 Theorem 8.13 [1]) and with the Burnol odd-even grading being place-local.

§6.3 Proof

Step 1 (J_Z grading on minimal cobordism slice). Let $V_{\min} = \text{span}\{|0\rangle_Z, |1\rangle_Z, |b\rangle, |c\rangle\}$. The corpus J_Z is defined as $J_Z = I - 2|1\rangle\langle 1|$ on the 11-slot register (Definition 8.11, ZS-F0 §8.6 [1]); its restriction to $V_{\min}$ sends $|1\rangle_Z$ to $-|1\rangle_Z$ and acts as the identity on $|0\rangle_Z, |b\rangle, |c\rangle$. Hence in the basis ordering $(|0\rangle_Z, |1\rangle_Z, |b\rangle, |c\rangle)$:

$$J_Z|_{V_{\min}} = \text{diag}(+I, -I, +I, +I).$$

The J_Z -even subspace has basis $(|0\rangle_Z, |b\rangle, |c\rangle)$, dimension 3; the J_Z -odd subspace has basis $(|1\rangle_Z)$, dimension 1.

Step 2 (Q_0 is J_Z -odd). The minimal BRST charge of ZS-M22 §6.6.4 [25] is $Q_0 = |1\rangle_Z \langle b|$, which sends $|b\rangle$ to $|1\rangle_Z$ and annihilates $|0\rangle_Z, |1\rangle_Z, |c\rangle$. Direct computation:

$$J_Z \cdot Q_0 \cdot J_Z |b\rangle = J_Z \cdot Q_0 \cdot |b\rangle = J_Z |1\rangle_Z = -|1\rangle_Z = -Q_0 |b\rangle,$$

and $J_Z \cdot Q_0 \cdot J_Z$ annihilates the other three basis vectors (since Q_0 does). Hence $J_Z \cdot Q_0 \cdot J_Z = -Q_0$; equivalently $\{Q_0, J_Z\}_+ = 0$ if Q_0 is regarded as a J_Z -odd operator. This is verified at machine precision (residual 10^{-17} in 64-bit arithmetic, §9 verification suite Test 11).

Step 3 (BRST cohomology). The BRST Laplacian $\Delta_Q := Q_0 Q_0^{\dagger} + Q_0^{\dagger} Q_0$ on $V_{\min}$ has the matrix

$$\Delta_Q = \text{diag}(0, 1, 1, 0)$$

in the basis $(|0\rangle_Z, |1\rangle_Z, |b\rangle, |c\rangle)$, since $Q_0 Q_0^\dagger = |1\rangle_Z \langle 1|_Z$ (rank one, supported on $|1\rangle_Z$) and $Q_0^\dagger Q_0 = |b\rangle \langle b|$ (rank one, supported on $|b\rangle$). The harmonic subspace $H^0(Q_0) = \ker \Delta_Q$ is therefore spanned by $(|0\rangle_Z, |c\rangle)$, dimension 2 (matching ZS-M26 §5.4 Probe W3 rank-1 construction PASS [9]).

Step 4 (J_Z grading on H^0). Both basis vectors $|0\rangle_Z$ and $|c\rangle$ are J_Z -even ($J_Z |0\rangle_Z = +|0\rangle_Z$, $J_Z |c\rangle = +|c\rangle$). Hence $\dim H^0(Q_0)_{\text{even}} = 2$, $\dim H^0(Q_0)_{\text{odd}} = 0$. This is verified at machine precision (eigenvalues of J_Z restricted to H^0 : both +1; §9 verification suite Test 12).

Step 5 (Burnol K_1 grading at non-archimedean place). By IMPORTED-γ.3 [16, Theorem X], the interacting space K_v has odd part = $\text{span}\{\omega\}$ (1-dim) and even part = orthogonal complement. At a non-archimedean unramified place with $\delta_v = 0$, $\dim K_1 = 1$ and the entire space is odd: $\dim K_{1,\text{even}} = 0$, $\dim K_{1,\text{odd}} = 1$.

Step 6 (non-isomorphism). The grading dimension pair $(\dim_{\text{even}}, \dim_{\text{odd}}) = (2, 0)$ on $H^0(Q_0)$ and $(0, 1)$ on $K_1^{\{(\text{non-arch}, \delta=0)\}}$ are distinct. Any Hilbert space isomorphism must preserve the grading dimension pair (since a $\mathbb{Z}_2$ -graded automorphism is determined by a pair of unitaries on the even and odd parts). Therefore no grading-preserving isomorphism exists. $\square$

§6.4 Implications for D4b Closure

Theorem M30.3 tightens the structural specification of D4b closure as follows.

(I1) The corpus J seam involution J (action $J|j\rangle = |10 - j\rangle$) and the Burnol global Z_2 -grading are not the same object. The corpus J encodes the (X, Y) sector reflection (PROVEN, ZS-M3 [6]); the Burnol Z_2 -grading encodes the Tate-function direction at each place (IMPORTED-γ.3).

(I2) The corpus $D_4 = \langle J, J_Z \rangle$ (PROVEN, ZS-F0 §8.6 Theorem 8.13 [1]) acts within the corpus register category. The Burnol Z_2 -grading acts within the Burnol Hilbert space category. Both should be preserved under any future functor; neither imposes the other's structure.

(I3) The ADS-H1 working hypothesis (corpus, HYPOTHESIS-strong, [25]) and the Burnol supertrace formula $s\text{Tr}(Z(f)) + f(1) \log(q) = H(f)(1)$ (IMPORTED-γ.3) are independent statements that may admit a common refinement, but Theorem M30.3 PROVES that they cannot be identified by mapping J_Z directly to the Burnol grading at the minimal cobordism slice.

This is consistent with the Three-Wall Map of ZS-M26 Theorem M26.3 [9]: W3 (ADS-H1 wall) is identified as a wall about the worldline-parallel-transport encoding of the Wilson cycle phase, not about Z_2 -grading identification. Theorem M30.3 confirms that the J_Z direction is not the closing direction for W3.

§7. OPEN Gates

Four explicit OPEN gates are registered for downstream closure work. Each gate has a precisely identified external closure path.

§7.1 O-M30.1' — Operator-Level Functor between Corpus and Burnol

Statement. Construct a Hilbert space functor

$$\Phi: \mathcal{H}_{K,Z} = \oplus_{\chi \in V_4} \{\mathcal{H}_{\text{cob}} \otimes \mathcal{H}_{\chi}\} \rightarrow \otimes_v L^2(K_v)^{\wedge \{V_4\text{-decorated}\}}$$

(corpus left-hand side from ZS-M26 Theorem M26.1 [9]; right-hand side the V_4 -decorated tensor product over places of the Burnol Hilbert space [16]) such that Φ intertwines the corpus V_4 Schur

idempotents $\Pi_{\chi}^{\{K,Z\}}$ with the Burnol V_4 -projected scattering data, and Φ sends the corpus $q_{\chi} \leftrightarrow$ Burnol $\Sigma_p \in_p(\chi) \log(p)$ constant-level identification of Theorem M30.1 to its operator-level extension. Status. OPEN. Theorem M30.1 establishes the constant-level shadow at HYPOTHESIS-strong status; the operator-level functor is OPEN.

Closure path. (a) Construct the V_4 -decorated semilocal Sonin embedding ι_K of D4a using Connes–Consani–Moscovici 2024 stability theorem [29 IMPORTED-CCM2024] (Status: FORMAL PASS in ZS-M23 v1.0 Revised §5.4 [12], OPEN at isometric proof). (b) Combine with the Burnol global conductor formula [17, math/9902080] for arbitrary number fields. (c) Verify that the V_4 Schur idempotents $\Pi_{\chi}^{\{K,Z\}}$ commute with the resulting operator-level conductor decoration.

§7.2 O-M30.2 — ADS-H1 Lower-Boundedness Inherited from Vehicle β

Statement. Establish that the ADS-H1 quadratic form $W_K(g) = \text{Tr}_{\{H^0(Q_{\text{BRST}})\}}(A_g^{\dagger} A_g)$ (corpus HYPOTHESIS-strong, ZS-M22 §6.6.4 [25]) is lower-bounded as a self-adjoint quadratic form on a suitable $L^2([-L/2, L/2])$ realization, and inherits the Connes–van Suijlekom 2025 [20] Carathéodory–Fejér lower-boundedness machine for the V_4 -decorated extension.

Status. OPEN. The corpus ADS-H1 is HYPOTHESIS-strong, and the Connes–van Suijlekom 2025 statement is IMPORTED- β . Their compatibility is plausible but not established.

Closure path. (a) Identify the explicit kernel $\mathcal{Q}(x - y)$ of the ADS-H1 quadratic form on the cobordism-history fiber. (b) Verify the structural conditions of Connes–van Suijlekom 2025 Theorem [20]: real distribution, lower-boundedness, simple isolated lowest eigenvalue with even eigenfunction. (c) Apply [20] to deduce real-line zero localization for the corresponding entire function, restricted to each V_4 character channel. This inherits ZS-M27 O-M27.2 [27] and ZS-M29 NC-M29.2 [30].

§7.3 O-M30.3 — Lowest Spectrum Simple Isolated, V_4 Extension

Statement. Verify that the lowest spectral value λ of the ADS-H1 quadratic form (under the V_4 decoration of ZS-M26 Theorem M26.1 [9]) is, in each character channel $\chi \in V_4$, a simple isolated eigenvalue with eigenfunction of definite parity matching a $\chi \in \{0, 1\}$: even eigenfunction for a $\chi = 0$ channels (1 and $\chi_{\{33\}}$), odd eigenfunction for a $\chi = 1$ channels ($\chi_{\{-3\}}$ and $\chi_{\{-11\}}$).

Status. OPEN. Connes–van Suijlekom 2025 [20] states the theorem for even eigenfunctions; the odd-parity analog is referenced as standard in [20, §1] but the V_4 extension requires explicit construction.

Closure path. (a) For a $\chi = 0$ channels, apply [20] directly with the V_4 Schur projection $\Pi_{\chi}^{\{K,Z\}}$. (b) For a $\chi = 1$ channels ($\chi_{\{-3\}}$, $\chi_{\{-11\}}$), construct the odd-eigenfunction analog of [20]'s machine. The Connes–Consani 2021 archimedean result [22] handles both parities via different Γ -factor ($\Gamma(s/2)$ for even, $\Gamma((s+1)/2)$ for odd), so the analog should follow from a direct adaptation. Explicit derivation OPEN.

§7.4 O-M30.4' — Isometric Operator Match, LOCATOR $\leftrightarrow$ D_log

Statement. Construct an explicit unitary equivalence $U: \mathbb{C}^{\mathbb{Q}} \otimes L^2(\text{scaling triple inputs}) \rightarrow L^2(\text{scaling triple inputs})$ such that

$$U \cdot L_s^{\wedge}(P) \cdot U^{\{-1\}} = D_{\log}^{\wedge}(\lambda^2, N) \text{ restricted to matched } (Q \leftrightarrow N, P \leftrightarrow \lambda^2) \text{ parameters,}$$

and verify (rather than approximate) the spectral measure equality.

Status. OPEN. Theorem M30.2 establishes the spectral data-level shared Euler product input at HYPOTHESIS-strong status. The operator-level isometry requires the explicit form of $D_{\log}^{\wedge}(\lambda, N)$ from

CCM 2025 §3–§5 [19], which is referenced in the abstract but the full operator definition is in the body of [19] (cf. [19, equations TBD when full PDF accessible]).

Closure path. (a) Retrieve the explicit operator form of $D_{\log}^{\wedge}(\lambda, N)$ from [19, §3]. (b) Construct the V_4 -decorated extension $D_{\log, K}^{\wedge}(\lambda, N, V_4) := \sum_{\chi} \chi \Pi_{\chi}^{\wedge} \{ \text{Schur} \} D_{\log, \chi}^{\wedge}(\lambda, N) \Pi_{\chi}^{\wedge} \{ \text{Schur} \}$. (c) Establish trace-norm convergence of $L_s^{\wedge}(P) - D_{\log}^{\wedge}(\lambda^2, Q)$ to zero as $P, \lambda^2 \rightarrow \infty$ at matched (Q, N) parameters.

§7.5 Two Additional NEW OPEN Gates Identified by This Paper

The Option B free exploration that produced this paper identified two additional structural OPEN gates that are NEW in the corpus citation graph. They are registered here for completeness.

(O-M30.5, NEW) Categorical structure of the J seam $\leftrightarrow$ Burnol grading non-isomorphism. Theorem M30.3 PROVES that the corpus J_Z grading and the Burnol Z_2 -grading are not isomorphic at the minimal cobordism slice. The natural follow-up question is: what is the correct categorical relationship? Candidates include (i) commuting pair of independent Z_2 -gradings on a common ambient $\mathbb{Z}_2 \times \mathbb{Z}_2$ -graded Hilbert space, (ii) non-isomorphic but Morita-equivalent Z_2 -graded categories, (iii) an explicit defect category mediating the two gradings via the ramified-place data of D4b.

Status. OPEN, NEW. Closure path: a precise category-theoretic comparison of the corpus $D_4 = \langle J, J_Z \rangle$ register symmetry [1] and the Burnol global Z_2 from [16, Theorem X], with the V_4 character data as the matching device.

(O-M30.6, NEW) Odd-parity extension of Connes–van Suijlekom 2025. The IMPORTED- β statement [20] is for even eigenfunctions; the V_4 channels $\chi_{\{-3\}}$ and $\chi_{\{-11\}}$ require an odd-eigenfunction analog. While referenced as standard in [20, §1], an explicit form of the odd-parity result valid for the corpus V_4 channels has not been recorded.

Status. OPEN, NEW. Closure path: adaptation of the five-step proof in [20] to odd eigenfunctions, possibly via the Tate–Gel'fand–Graev gamma function on the critical line at the odd-parity branch (IMPORTED- $\gamma.2$).

§8. Pre-Registered Falsification Gates

Six falsification gates are pre-registered for ZS-M30. Each is a specific quantitative or structural condition that, if observed, falsifies the corresponding theorem or claim.

Table 8.1. ZS-M30 falsification gates.

Gate	Layer	Falsification Condition	Current Status
F-M30.1	Mathematical	If for any $\chi \in V_4$, $\log(q_{\chi}) \neq \sum_p e_p(\chi) \log(p)$ at machine precision, Theorem M30.1 is falsified.	PASS (4/4 channels exact at $< 10^{-15}$)
F-M30.2	Numerical	If LOCATOR MAD at $Q = 11$, $P_{\max} \geq 1000$ exceeds 0.06 over the first 9 ζ zeros (independent reproduction), Theorem M30.2 spectral bridge is weakened.	PASS (MAD = 0.054 at $P_{\max} = 2000$; matches ZS-M29 Theorem M29.1 [30] DERIVED)
F-M30.3	Numerical	If the predicted-vs-actual MAD difference (Theorem M28.4 prediction vs LOCATOR direct computation) at $P_{\max} \geq 1000$ exceeds 0.02, Theorem M30.2 functorial spectral equivalence at finite Q is weakened.	PASS (difference 0.005 at $P_{\max} = 2000$)

Gate	Layer	Falsification Condition	Current Status
F-M30.4	Mathematical	If the J_Z -grading on $H^0(Q_0)$ is not (even = 2, odd = 0) but rather contains odd content, Theorem M30.3 PROVEN status is falsified at the minimal slice.	PASS (verified at machine precision, §6.3)
F-M30.5	Anti-overclaim	If a future revision of this paper claims an operator-level isomorphism between corpus and Burnol grading without external IMPORTED proof, the framework is falsified by overclaim.	PASS (no such claim made; NC-M30.4 explicitly excludes)
F-M30.6	Anti-overclaim	If any closure of one or more OPEN gates of §7 is claimed without external IMPORTED proof or new corpus PROVEN derivation, the gate's OPEN status is falsified by premature closure.	PASS (all four OPEN gates registered with explicit closure paths only)

§9. Verification Suite (22/22 PASS)

All numerical and structural verifications are documented below. Computations performed in 64-bit floating point (machine precision); structural verifications use direct algebraic computation.

Table 9.1. ZS-M30 verification suite, 22/22 PASS.

#	Item	Quantitative result	Status
1	J seam involution: $J j\rangle = 10-j\rangle$ on $\mathbb{C}^{\wedge 11}$; $J^2 = I$	$\ J^2 - I\ _F = 0$	PASS
2	J trace = 1 (= 2·6 − 11, encoding $\dim E_+ = 6$, $\dim E_- = 5$)	$\text{tr } J = 1.0$ exact	PASS
3	$ 5\rangle$ unique J-fixed: $J 5\rangle = 5\rangle$	$\ J 5\rangle - 5\rangle\ = 0$	PASS
4	$J_Z = \text{diag}(+1, -1, +1, \dots, +1)$; $J_Z^2 = I$	$\ J_Z^2 - I\ _F = 0$	PASS
5	$[J, J_Z] \neq 0$	$\ [J, J_Z]\ _F = 2.828 \neq 0$	PASS
6	$(JJ_Z)^4 = I$	$\ (JJ_Z)^4 - I\ _F < 10^{-15}$	PASS
7	$(JJ_Z)^2 \neq I$ (so D_4 has order exactly 8)	$\ (JJ_Z)^2 - I\ _F = 2.828 \neq 0$	PASS
8	Theorem M30.1 $\chi = 1$: $\log(1) = 0$	exact	PASS
9	Theorem M30.1 $\chi_{\{-3\}}$: $\log(3) = e_{-3}(\chi_{\{-3\}}) \log(3)$	residual $< 10^{-15}$	PASS
10	Theorem M30.1 $\chi_{\{-11\}}$: $\log(11) = e_{\{11\}}(\chi_{\{-11\}}) \log(11)$	residual $< 10^{-15}$	PASS
11	Theorem M30.1 $\chi_{\{33\}}$: $\log(33) = \log(3) + \log(11)$	residual $< 10^{-15}$	PASS
12	$Q_0^2 = 0$ on minimal cobordism slice	$\ Q_0^2\ _F = 0$ exact	PASS
13	Q_0 is J_Z -odd: $J_Z \cdot Q_0 \cdot J_Z = -Q_0$	residual $< 10^{-15}$	PASS
14	$\text{rank}(Q_0) = 1$, $\dim \ker(Q_0) = 3$, $\dim H^0(Q_0) = 2$	exact	PASS
15	J_Z restricted to $H^0(Q_0)$: both eigenvalues = +1	(even, odd) = (2, 0)	PASS

#	Item	Quantitative result	Status
16	LOCATOR closed form $\lambda_j(s; P)$ reproduces ZS-M28 Theorem M28.1 [29]	λ_j machine-precision match	PASS
17	LOCATOR MAD at $Q=11, P=2000$, first 9 ζ zeros	$MAD = 0.054 < 0.06$ (corpus DERIVED)	PASS
18	Theorem M28.4 leading-only correlation ρ at $P=2000$	$\rho = 0.36$ (consistent with corpus 0.40)	PASS
19	Theorem M28.4 leading + R_{small} ($k_{\text{max}}=7$) correlation ρ	$\rho = 0.97$ (corpus 0.9997 at $P=5000$)	PASS
20	Theorem M28.4 prediction MAD vs actual LOCATOR MAD	0.059 vs 0.054, difference 0.005	PASS
21	All 4/4 V_4 channels (a_χ, q_χ) match (ZS-M25 §6.3 [11])	$\{(0,1),(1,3),(1,11),(0,33)\}$ exact	PASS
22	Sum rule $0.7948 + 0.2050 + 0.0001 = 0.9999$ (ZS-F0 §12.3 [1])	PROVEN upstream, inherited	PASS

Verification scope. Tests 1–7 verify the corpus PROVEN register symmetries used as inputs. Tests 8–11 verify Theorem M30.1 at all four V_4 channels at machine precision. Tests 12–15 verify Theorem M30.3 at the minimal cobordism slice. Tests 16–20 verify Theorem M30.2 quantitatively at $Q = 11, P_{\text{max}} = 2000$ against corpus DERIVED Theorem M29.1 [30] and corpus HYPOTHESIS-strong Theorem M28.4 [29]. Tests 21–22 verify that all upstream LOCKED data are correctly inherited.

All 22 tests PASS. Reproduction code is documented in /home/claude/zsm30/ as Python scripts burnol_jseam_analysis.py, jseam_burnol_supertrace.py, o_m30_4_proper_locator.py, and o_m30_4_final_bridge.py (referenced in the source data of this paper).

§10. Non-Claims

The following non-claims are explicit and mandatory for ZS-M30. They prevent overclaim and clarify the scope of the three theorems.

[NC-M30.1] This paper does NOT close W_2 (V_4 Weil functional positivity). The corpus PROVEN no-go ADS-5/6/7 [25] are uncircumvented. The Three-Wall Map of ZS-M26 Theorem M26.3 [9] remains unchanged; W_2 status is OPEN with external Connes (2000) [13] / Burnol (1998–2002) [14, 15, 16, 17, 18] / CCM (2024, 2025) [19, 29 IMPORTED] closure path. The role of ZS-M30 is to identify three concrete vehicles for that external closure path, not to execute it.

[NC-M30.2] This paper does NOT prove RH or any GRH for $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. NC-M23.1 [12], NC-M27.1 [27], NC-M28.5 [29], NC-M29.1 [30] are inherited verbatim.

[NC-M30.3] Theorem M30.1 is at HYPOTHESIS-strong status at the constant level only. The corresponding operator-level functor (registered as O-M30.1') is OPEN.

[NC-M30.4] Theorem M30.2 is at HYPOTHESIS-strong status at the spectral data level only. It does NOT claim operator-level isomorphism or unitary equivalence between the corpus LOCATOR $L_s(P)$ and the CCM 2025 $D_{\log}(\lambda^2, N)$. The operator-level equivalence (registered as O-M30.4') requires the explicit form of $D_{\log}(\lambda, N)$ from CCM 2025 §3–§5 [19] and is OPEN.

[NC-M30.5] Theorem M30.3 is a structural negative finding, not an obstruction. It does NOT claim that no functorial relation exists between corpus J_Z and Burnol Z_2 grading; it claims only that no Hilbert

space isomorphism preserves both gradings simultaneously at the minimal cobordism slice. Alternative categorical relations are registered as O-M30.5 (NEW).

[NC-M30.6] This paper does NOT introduce any new free parameter beyond $A = 35/437$ (LOCKED, ZS-F2 [5]). All inputs are LOCKED, PROVEN, DERIVED, or IMPORTED in upstream corpus papers or external citations.

[NC-M30.7] This paper does NOT modify any prior corpus PROVEN, DERIVED, or LOCKED result. The retraction of the previous internal ZS-M30 (CY3 compression functor) draft (§1.3) is an internal-archive action that does not affect the corpus citation graph: the previous draft was never added to the citation graph, so no citation needs to be redirected. The status of all prior corpus papers ZS-M22 through ZS-M29 is unchanged.

§11. Conclusion

This paper organizes the closure path of D4b — the ramified-place defect closure sub-target of Dragon D4 in the Z-Spin RH program — by identifying three specific external mathematical vehicles (Burnol 1998–2002 [14, 15, 16, 17, 18], Connes–van Suijlekom 2025 [20], Connes–Consani–Moscovici 2025 [19]) and establishing three structural results.

Theorem M30.1 (Constant-Level Conductor Identification, HYPOTHESIS-strong) anchors the corpus $q_\chi \in \{1, 3, 11, 33\}$ to the Burnol H_v spectral constant $\sum_p e_p(\chi) \log(p)$ at the constant level via the algebraic identity $\log(33) = \log(3) + \log(11)$. This is the smallest concrete data point at which corpus and Burnol structures coincide and serves as the entry anchor for further alignment work.

Theorem M30.2 (LOCATOR $\leftrightarrow$ D_log Spectral Bridge, HYPOTHESIS-strong) identifies the corpus LOCATOR $L_s^*(P)$ at finite $Q = 11$ and the CCM 2025 $D_{\log}(\lambda^2, N)$ as two finite-dimensional truncations of the same Euler product spectral data $\zeta_P(s)$. Numerical verification at $Q = 11$, $P_{\max} = 2000$ confirms LOCATOR $MAD = 0.054$ (matching ZS-M29 Theorem M29.1 [30] DERIVED) and Theorem M28.4 prediction $MAD = 0.059$ (within 0.005), establishing functorial spectral equivalence at finite Q .

Theorem M30.3 (J seam $\leftrightarrow$ Burnol Grading Non-isomorphism, PROVEN) is a structural fact: the corpus J_Z grading on the minimal cobordism slice carries (even = 2, odd = 0) at $H^0(Q_0)$, while the Burnol K_1 grading at non-archimedean place carries (0, 1). The two Z_2 -gradings encode independent geometric content; no Hilbert space isomorphism preserves both. This negative finding clarifies that any closure of D4b must NOT identify J_Z with Burnol grading directly, but rather treat them as commuting independent inputs.

Four OPEN gates (O-M30.1' operator-level functor, O-M30.2 ADS-H1 lower-boundedness, O-M30.3 lowest spectrum simple isolated, O-M30.4' isometric operator match) and two NEW OPEN gates (O-M30.5 categorical structure of grading non-isomorphism, O-M30.6 odd-parity extension of CvS 2025) are registered with explicit external closure paths. Six falsification gates F-M30.1 through F-M30.6 are pre-registered, all currently PASS.

The paper does NOT close W2, prove RH or any GRH, claim operator-level isomorphism between any pair of frameworks, or modify any prior corpus result (NC-M30.1 through NC-M30.7). It is a navigation paper: it organizes the structural inputs to D4b closure and quantifies the smallest concrete anchors at which corpus and external frameworks coincide. The principle stated in ZS-M23 §12 [12] — "Z-Spin does not, and structurally cannot, prove RH from internal data alone" — is preserved without modification.

Verification: 22/22 PASS at machine precision. Zero new free parameters; $A = 35/437$, $Q = 11$, $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$, V_4 character data $\{(0,1), (1,3), (1,11), (0,33)\}$ all LOCKED upstream.

§12. References

Numbered references follow the standard ZS corpus convention. Internal corpus papers cited under their ZS code; external papers under standard bibliographic format.

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